

EE2080: Random Processes

Lecture1: Introduction

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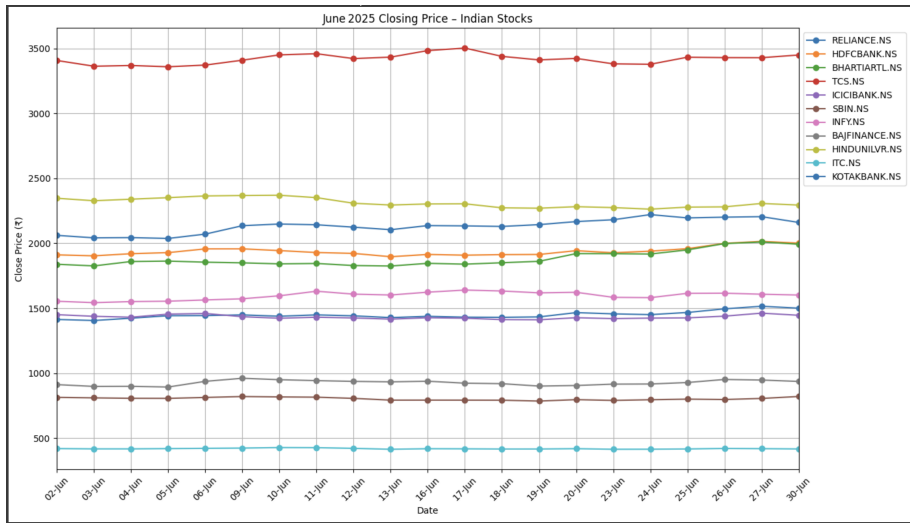
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Definition

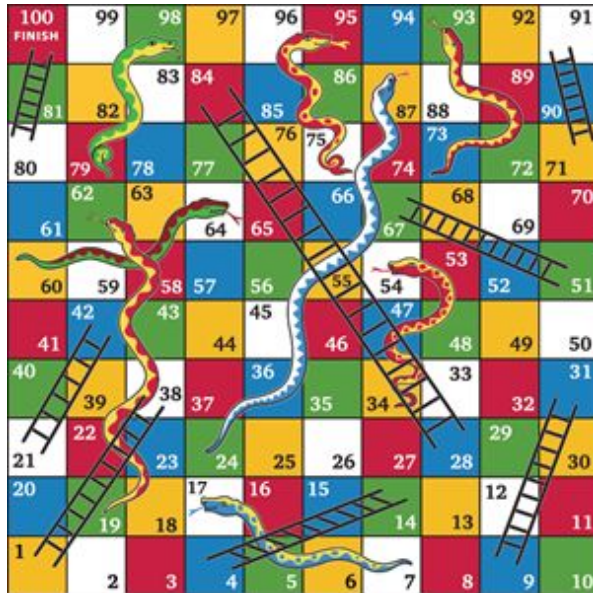
A random process is an indexed collection $X : \{X_t\}_{t \in T}$ of random variables all defined w.r.t the same probability space.

- The index set T is often interpreted as time.
- If T is finite, X is a random vector
- If T is countably infinite: Discrete Random Process
- If T is uncountably infinite: Continuous Random Process

Example 1: Stock Market Fluctuations

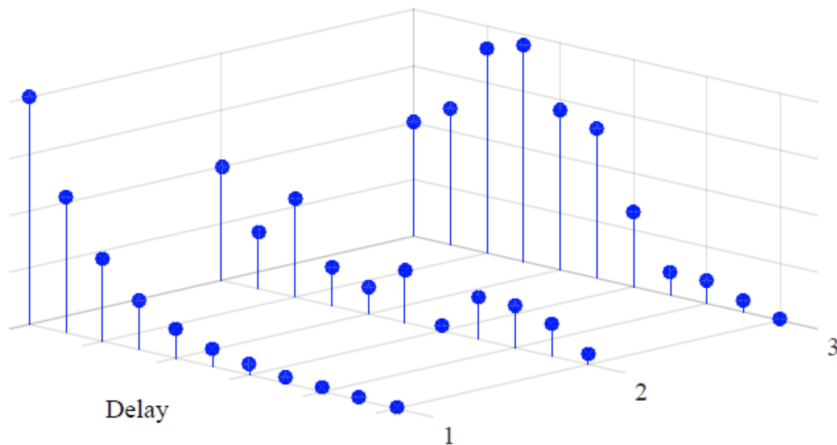


Example 2: Snake and Ladder Game



Example 3: PageRank

Example 4: Channel Impulse Response of a Time-Varying Multi-path Wireless Channel



Course Logistics